Value-at-Risk Forecasting Mini Program for SPY Daily Returns

1. Introduction and Task

This project forecasts one-day-ahead Value-at-Risk (VaR) for SPY daily log returns at the 1%, 5%, and 10% lower-tail levels. The final project is organized as a clean mini program rather than a claim of a new model contribution. The main workflow connects data exploration, rolling-window VaR estimation, statistical backtesting, and model comparison.

The retained main models are Historical Simulation, KDE-weighted Historical Simulation, GARCH-t, GJR-GARCH-t, and direct MLP quantile regression.

2. Data and Forecasting Framework

The data set contains 4,640 daily observations for SPY from 2000-01-04 to 2018-06-27. The target variable is daily log return. The evidence shows heavy tails and volatility clustering, which motivates both empirical-tail methods and conditional-volatility models.

VaR is treated as a lower conditional quantile. For tail probability alpha, a correct VaR forecast should be violated approximately alpha of the time. The final comparison uses 3640 common forecast dates across all retained main models, so the table is an apples-to-apples comparison.

3. Model Design

Historical Simulation is transparent and distribution-free, but it reacts slowly when volatility regimes change. KDE-weighted Historical Simulation smooths the empirical tail and gives more importance to recent returns. GARCH-t and GJR-GARCH-t provide interpretable time-varying volatility benchmarks with Student-t innovations. The MLP quantile-regression model is included as a neural benchmark trained with pinball loss, not as an assumed winner.

4. Main Backtesting Results

The aligned failure rates are:

- GARCH(1,1)-t: 0.0168, 0.0651, 0.1187 at the 1%, 5%, and 10% VaR levels.

- GJR-GARCH(1,1)-t: 0.0146, 0.0654, 0.1115 at the 1%, 5%, and 10% VaR levels.

- Gaussian KDE Weighted HS: 0.0129, 0.0459, 0.0898 at the 1%, 5%, and 10% VaR levels.

- MLP Quantile: 0.0047, 0.0332, 0.2115 at the 1%, 5%, and 10% VaR levels.

- Weighted Historical Simulation: 0.0168, 0.0530, 0.1019 at the 1%, 5%, and 10% VaR levels.

The lowest pinball losses on the aligned sample are: 1% VaR - GJR-GARCH(1,1)-t (0.000305); 5% VaR - GJR-GARCH(1,1)-t (0.001150); 10% VaR - GJR-GARCH(1,1)-t (0.001874). These loss results should be interpreted together with coverage and independence diagnostics.

5. Comparative Interpretation

The results support a conservative interpretation. Historical methods are easy to audit but can struggle during regime changes. GARCH-type models add useful volatility structure but do not perfectly solve unconditional coverage. The direct MLP benchmark is flexible, but its tail behaviour is unstable under limited rolling-window tail observations. Therefore, the main contribution of the mini program is a reproducible and statistically disciplined VaR comparison, not a claim that neural methods automatically dominate classical risk models.

6. Appendix: Exploratory Model C Extension

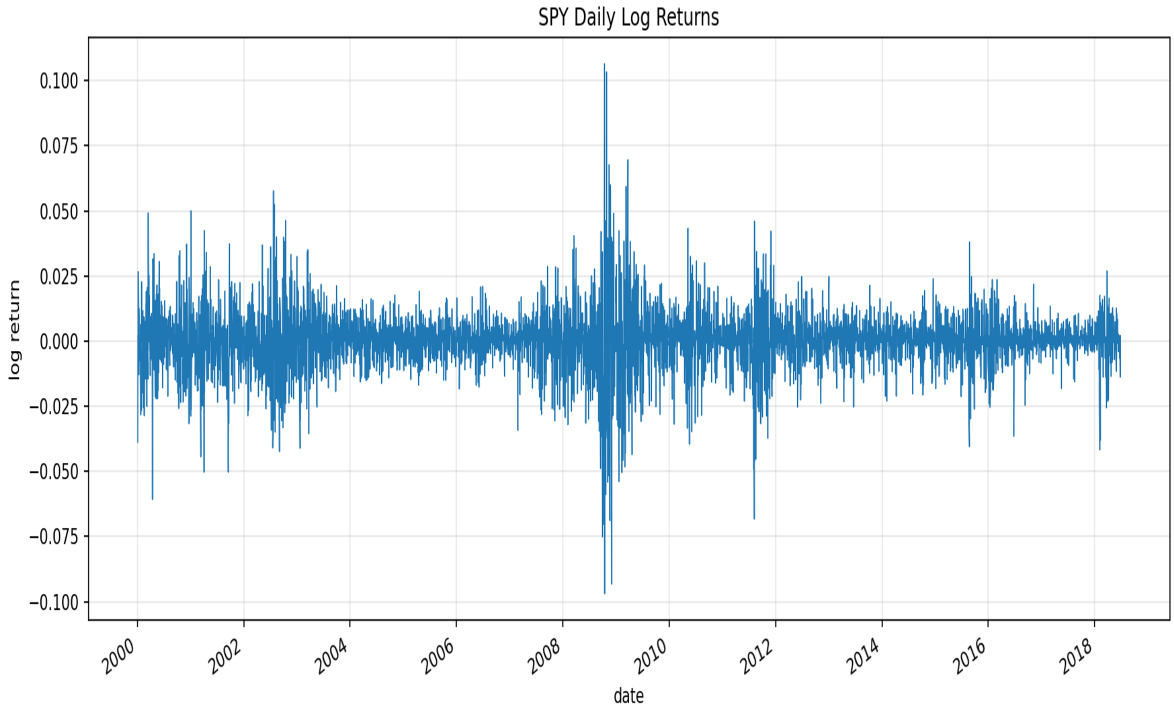
I also preserve an exploratory Model C extension in the archive and discuss it here as technical evidence, not as the main result. Model C uses a GARCH-based risk anchor and a neural correction layer. It is interesting because it shows how a neural model can be constrained by financial risk structure rather than asked to learn the full VaR level from scratch.

This extension should be read with two caveats. First, its evaluation sample is shorter than the main aligned comparison. Second, some regularization choices were selected after inspecting out-of-sample failure-rate behaviour. For that reason, Model C is presented as an exploratory extension and not as the primary ranking basis.

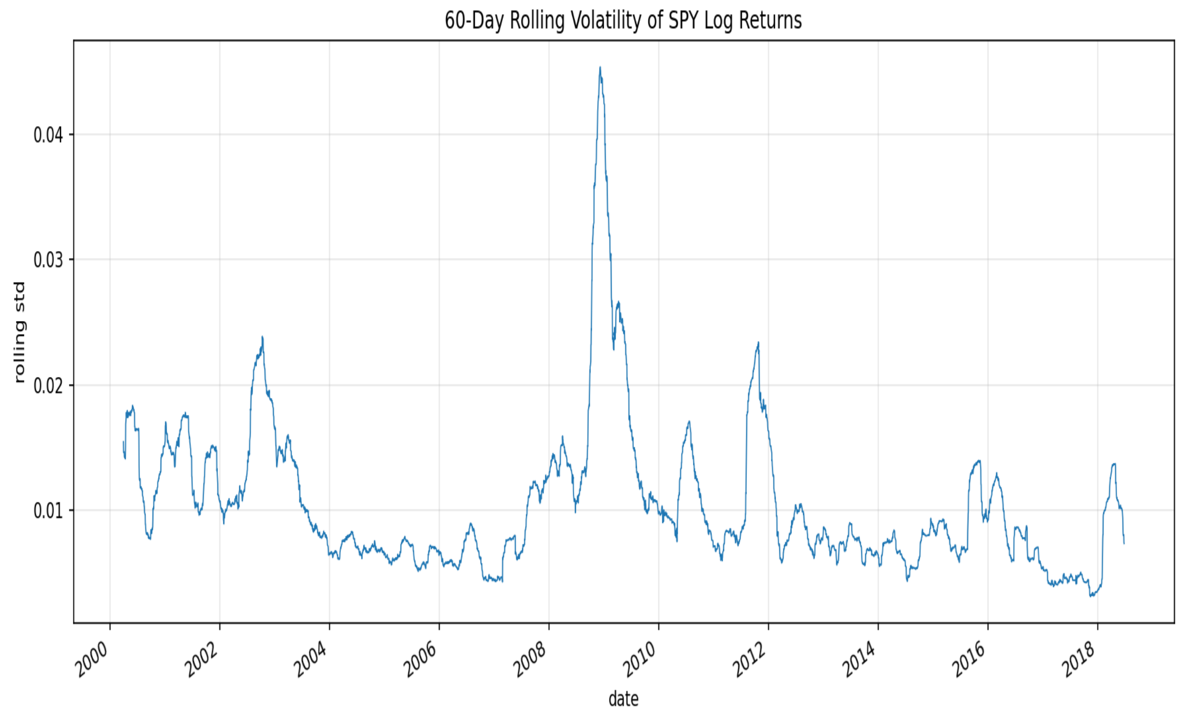
7. Conclusion

This unified project demonstrates statistical modelling, Python implementation, financial risk understanding, and critical comparison. The final submission is deliberately concise and reproducible, while the archive preserves the deeper experimental work for reference.

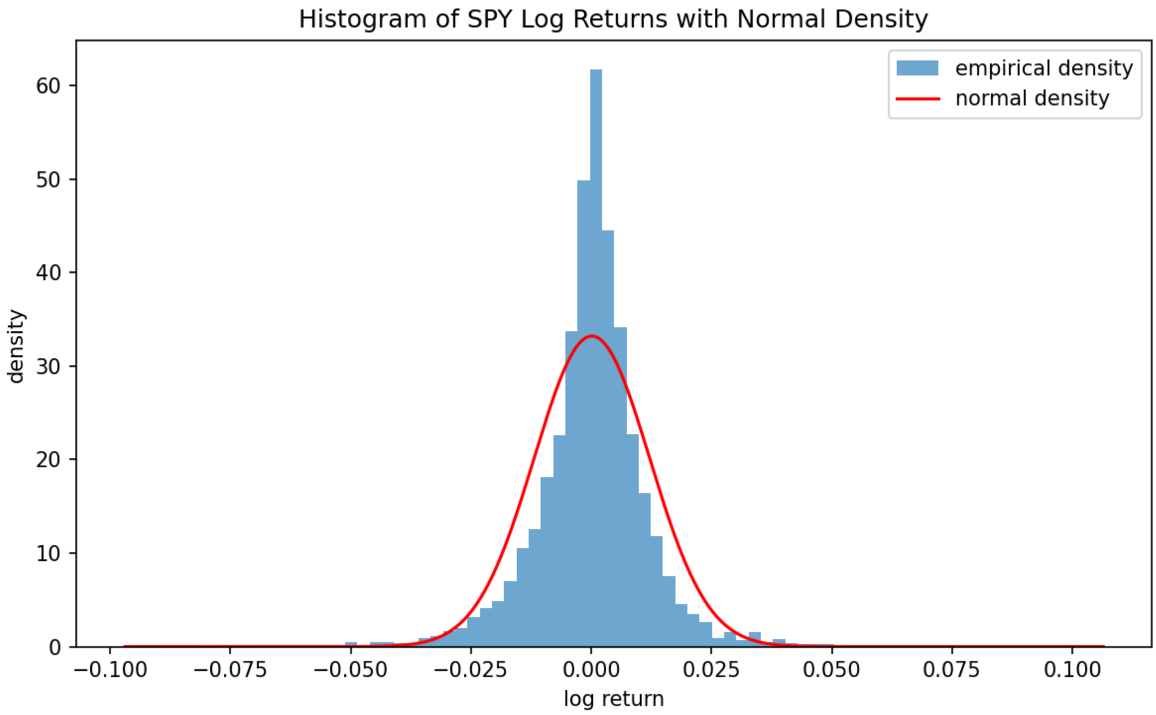
Return Series



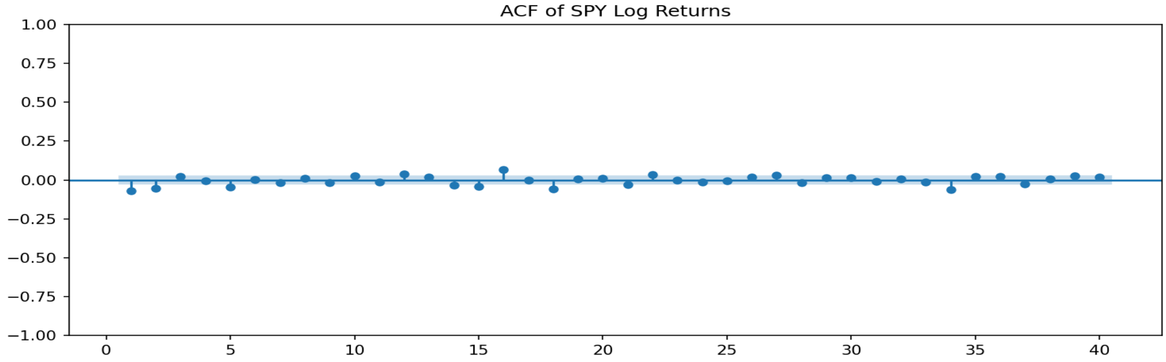
Rolling Volatility

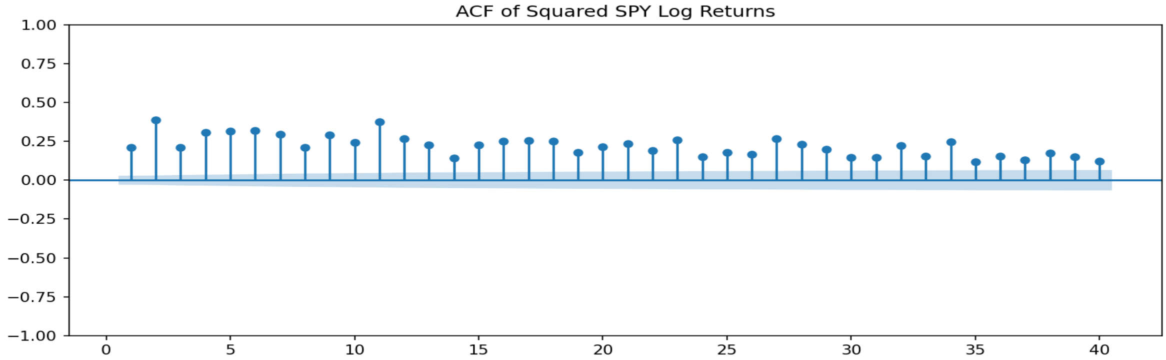


Return Distribution

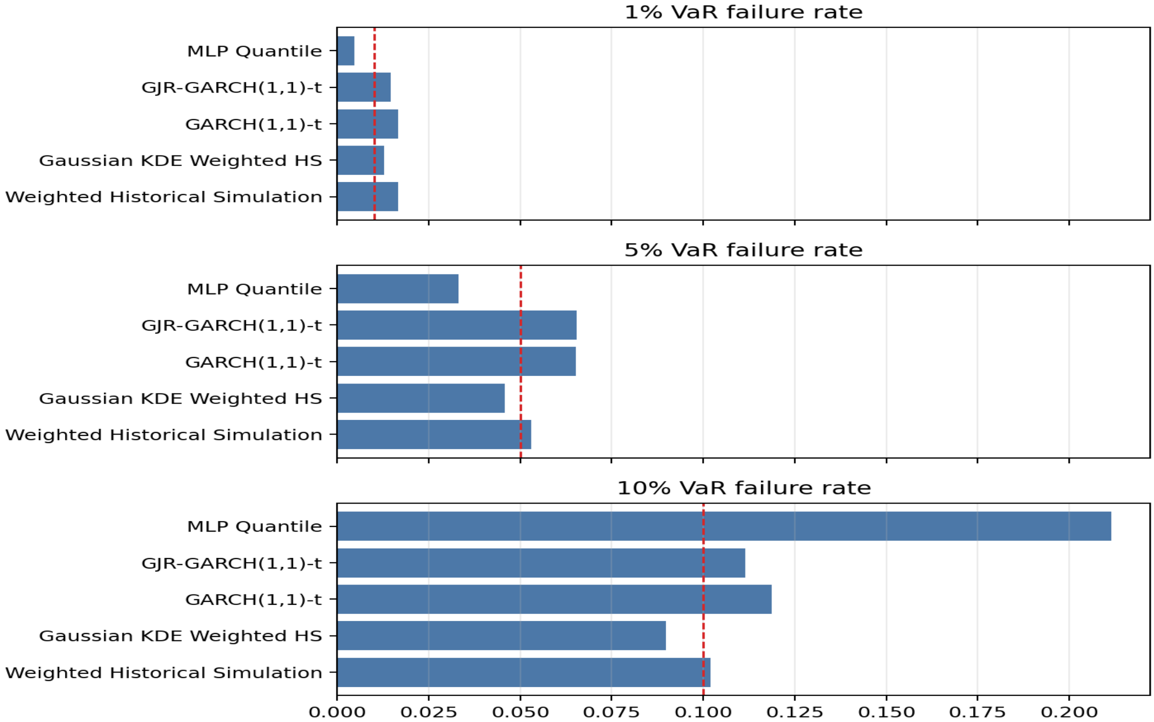


Autocorrelation Diagnostics





Aligned Failure Rates



Aligned Main Backtesting Table

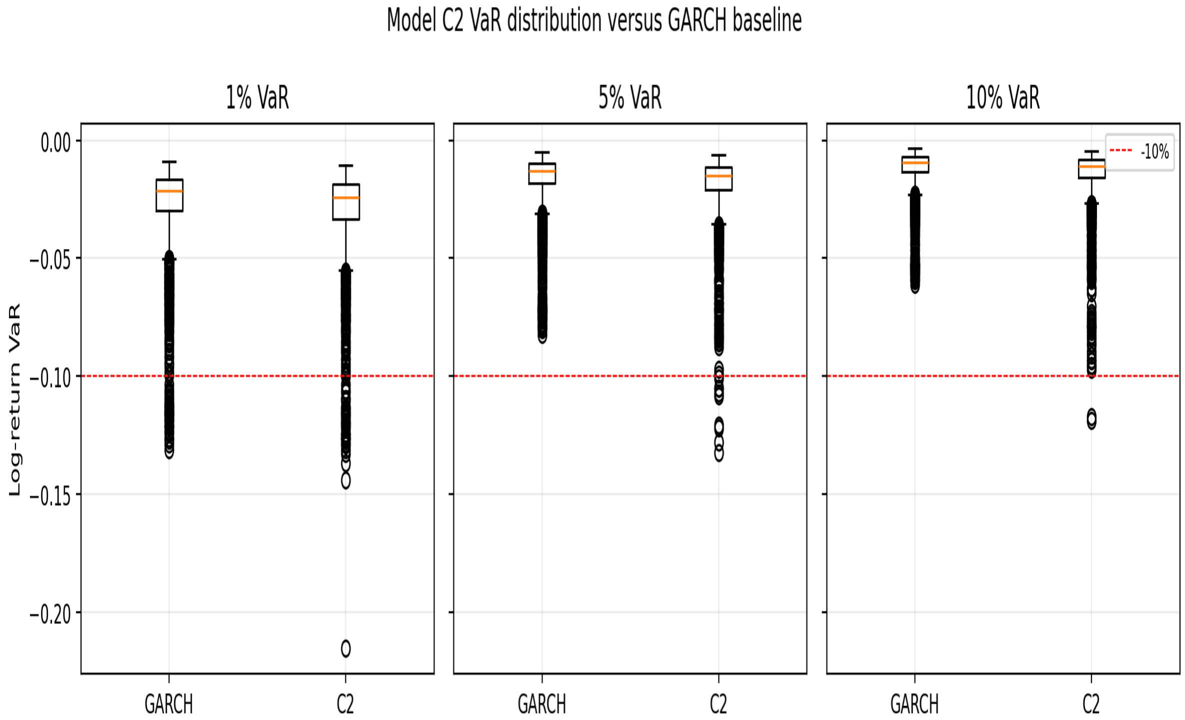
Model	Alpha	Viol./Exp.	Fail. rate	Kupiec p	Christoffersen p	Pinball
Weighted Historical Simulation	1%	61/36.4	0.0168	0.0002	0.0218	0.000362
Weighted Historical Simulation	5%	193/182.0	0.0530	0.4072	0.2366	0.001194
Weighted Historical Simulation	10%	371/364.0	0.1019	0.6998	0.3569	0.001925
Gaussian KDE Weighted HS	1%	47/36.4	0.0129	0.0911	0.0246	0.000351
Gaussian KDE Weighted HS	5%	167/182.0	0.0459	0.2476	0.0638	0.001198
Gaussian KDE Weighted HS	10%	327/364.0	0.0898	0.0379	0.1349	0.001921
GARCH(1,1)-t	1%	61/36.4	0.0168	0.0002	0.1045	0.000323
GARCH(1,1)-t	5%	237/182.0	0.0651	<0.0001	0.6919	0.001177
GARCH(1,1)-t	10%	432/364.0	0.1187	0.0003	0.1858	0.001899
GJR-GARCH(1,1)-t	1%	53/36.4	0.0146	0.0096	0.8009	0.000305
GJR-GARCH(1,1)-t	5%	238/182.0	0.0654	<0.0001	0.9066	0.001150
GJR-GARCH(1,1)-t	10%	406/364.0	0.1115	0.0224	0.0784	0.001874
MLP Quantile	1%	17/36.4	0.0047	0.0003	0.6895	0.001466
MLP Quantile	5%	121/182.0	0.0332	<0.0001	0.0087	0.003867
MLP Quantile	10%	770/364.0	0.2115	<0.0001	<0.0001	0.008696

Appendix: Exploratory Model C Results

Model C is shown as an exploratory extension only. It is not used in the main model ranking because its evaluation window and tuning protocol differ from the aligned main comparison.

Model	Alpha	Viol./Exp.	Failure rate	Kupiec p	Christoffersen p	Pinball loss
Model C1	1%	104 / 26.40	0.0394	0.0000	0.1721	0.000603
Model C1	5%	287 / 132.00	0.1087	0.0000	0.2463	0.001621
Model C1	10%	394 / 264.00	0.1492	0.0000	0.8393	0.002418
Model C2	1%	25 / 26.40	0.0095	0.7823	0.0217	0.000350
Model C2	5%	131 / 132.00	0.0496	0.9288	0.5196	0.001266
Model C2	10%	235 / 264.00	0.0890	0.0557	0.4737	0.002069

Model C VaR Distribution Diagnostic



Model C Correction Distribution